

Lab Sheet 1

1. Create variables 'name' and 'batch'. Next, assign 'Amrita Vishwa Vidyapeetham' and '2025' to those variables. Write the code for the following output:

My college's name is Amrita Vishwa Vidyapeetham, and my batch is 2025.

2. Get your name and age using the input function and print the following output:

Your name is _____ and your age is _____

3. Create a variable 'rupees', assign 4500 and get the output: *I don't have 4500 rupees in my wallet*. Next, assign 3000 to the same variable and then print: *But I have "3000" rupees*.
4. Write a code for printing the roots of $x^2 + 4x + 3 = 0$.
5. Write a program that will convert degrees Celsius to degrees Fahrenheit.
6. Write a program to find the sum of n natural numbers. (n will be input)
7. Write a Python code to print the speed if a runner completes 750 m in two and a half minutes.
8. Write Python code for the following programs; use an input function to get the needed values:
 - a. Adding two numbers.
 - b. Multiplying two numbers.
 - c. Area of the square.
 - d. Area of the rectangle.
 - e. Area of the circle.
 - f. The average of four numbers.
 - g. Calculate the arc length. (Length of an Arc = $\theta \times r$, where θ is in radians. Length of an Arc = $\theta \times (\pi/180) \times r$, where θ is in degrees.)
 - h. Calculate the surface volume and area of a sphere. (Surface area of the sphere = $4\pi r^2$. Volume of a sphere = $4/3 \times \pi r^3$.)
 - i. surface volume and area of a cylinder. (Surface area of a cylinder is $A = 2\pi rh + 2\pi r^2$, where r = radius, h = height. Volume of a cylinder is $V = \pi r^2 h$, where r = radius, h = height.)
9. Calculate the expressions:
 - a. $2.3 \left(3x + \frac{6y+z^2}{3} \right)$. Get x, y, z using the input function.
 - b. $\sqrt{5}((a^3)^{b^2} + 2ac - c)$. Get a, b, c using the input function.
10. Define 2 complex numbers x and y. Write Python programs for the following:
 - a. Add, subtract, multiply, and divide two complex numbers.
 - b. Print the complex conjugate, real and imaginary parts of a complex number.
11. Write a Python program to convert rectangular coordinates to polar coordinates in the first quadrant. (The polar coordinates of (a, b) is (r, θ) , where $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}(b/a)$)
12. Write a program to find the following expression. (x is given as input)
 - (a) $\cos(\sin x^2)$
 - (b) $\sec(\cot(\sqrt{e^x}))$
 - (c) $\log_{10}(\sin 2x)$
 - (d) $\cos^{-1}(x) + \log_5(x + 1)$